

tests/dynamic/suites/ stress_suite.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Authored for P10. Honest-SKIPs (topology_unsupported, §11.4.69)
until the live dynamic stack lands; parse-clean + authored today.

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

§11.4.85 / §11.4.169 **stress** suite for the dynamic data plane:
sustained-load

- concurrent-contention against the live dynamic stack (Squid + external acl helper + gluetun + Redis). It runs $N \geq 100$ sequential requests + $N \geq 10$ concurrent requests, captures each request's `%{http_code}` to its **own file** (so a verdict never rests on a job exit status), and records the latency/200-ratio. Every PASS cites a captured-evidence artefact — no metadata-only PASS (§11.4.69).

Prerequisites

- Source library tests/dynamic/lib/analyzer_common.sh + committed tests/lib/evidence.sh.
- curl, POSIX sh, date.
- For a real (non-SKIP) run: HELIX_DYNAMIC_STACK=1 + HELIX_PROXY_URL.
- Write access to qa-results/ (gitignored).

Usage examples

```
# Today (no live stack): honest SKIP, exit 0:  
bash tests/dynamic/suites/stress_suite.sh
```

```
# P10 live run with overrides:
```

```
HELIX_DYNAMIC_STACK=1 HELIX_PROXY_URL=http://127.0.0.1:34128 \
  STRESS_SEQ=100 STRESS_CONC=10 bash tests/dynamic/suites/
  stress_suite.sh
```

Env knobs: STRESS_SEQ (default 100), STRESS_CONC (default 10),
STRESS_TARGET (default http://target-a.internal/),
STRESS_OK_RATIO_PCT (min % of 200s for GREEN, default 95).

Edge cases

- **Live stack absent** → dyn_skip_if_no_stack emits topology_unsupported SKIP, prints the P10 note, exits 0.
- **200-ratio below STRESS_OK_RATIO_PCT** under sustained/concurrent load → FAIL citing stress.evidence.
- **Concurrent burst** uses one code-file per worker (the B3 bluff fix); a worker that errors writes 000, never relies on \$?.

§11.4.115 RED_MODE polarity

RED_MODE	What it asserts	PASS means
0 (default)	GREEN guard	stack sustains the load (ratio $\geq$ STRESS_OK_RATIO_PCT)
1	RED baseline	the pre-fix/throttled stack collapses (ratio < threshold) → defect reproduced

A RED_MODE=1 run where the stack does NOT collapse is a finding (defect not reproduced), not a pass.

Internal behaviour

- #!/usr/bin/env bash, POSIX-clean (sh -n + bash -n, §11.4.67).
- Sequential loop → seq.N.code; concurrent burst → conc.N.code (bounded by STRESS_CONC, nice); ratio + counts written to stress.evidence.
- GREEN requires both the sequential ratio and the concurrent-OK count to meet the threshold; otherwise FAIL. Evidence dir: qa-results/p9-harness/stress_<run-id>/.
- Resources: shell+curl only, capped concurrency, well under the §12.6 60% host-memory ceiling.

Related

- tests/dynamic/lib/analyzer_common.sh — sourced base (ab_pass_with_evidence, dyn_skip_if_no_stack, dyn_red_mode).

- tests/dynamic/suites/run_all_suites.sh — orchestrates this suite.
- Constitution §11.4.85 / §11.4.69 / §11.4.107 / §11.4.115 / §11.4.50 / §12.6; design spec §13.

Last verified

2026-07-01 — run with no live stack: honest SKIP (topology_unsupported), exit 0; sh -n + bash -n parse-clean. Live sustained/concurrent load is exercised in **P10**.